

Shapiro-Wilk Normality Test

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Introduction

The Shapiro-Wilk test, introduced by Shapiro and Wilk in 1965, is the most widely used and generally most powerful test of Normality for small-to-moderate sample sizes. It compares the observed sample order statistics with their expected values under Normality, producing a statistic W that is close to 1 when the sample is well approximated by a Normal distribution and substantially lower otherwise. Shapiro-Wilk is the default Normality test in R for samples between 3 and 5000 observations and is routinely used as a preliminary diagnostic before applying parametric inference (one- and two-sample t -tests, ANOVA, regression on the residuals) that nominally requires Normal data.

Prerequisites

A working understanding of order statistics, the Normal distribution, and the role of Normality assumptions in parametric inferential statistics.

Theory

For a sorted sample $x_{(1)} \leq \dots \leq x_{(n)}$, the Shapiro-Wilk statistic is

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

where a_i are tabulated constants derived from the expected values and covariance matrix of the order statistics of a standard Normal sample. The numerator is essentially the squared correlation between the sample order statistics and their expected values, scaled to produce $W \in (0, 1]$. Values near 1 indicate a Normal sample, lower values indicate departure; small W with correspondingly small p -value rejects the null of Normality.

Assumptions

Observations are independent and identically distributed; sample size lies within R's implementation range $n \in [3, 5000]$. The test does not assume the sample is Normal under the null — the null is precisely that hypothesis — but it does assume the observations are otherwise exchangeable.

R Implementation

```
set.seed(2026)

x_norm <- rnorm(60)
shapiro.test(x_norm)

x_skew <- rexp(60, rate = 1)
shapiro.test(x_skew)

x_t3 <- rt(60, df = 3)
shapiro.test(x_t3)
```

```
set.seed(2026)
pvals <- replicate(1000, shapiro.test(rt(20, df = 5)))$p.value
mean(pvals < 0.05)
```

Output & Results

The three samples illustrate the test's behaviour: a truly Normal sample of 60 observations gives $W = 0.98$, $p = 0.52$ (no rejection); an exponentially-distributed (right-skewed) sample of 60 gives $W = 0.83$, $p = 0.00002$ (strong rejection); a t_3 -distributed (heavy-tailed) sample gives $W = 0.93$, $p = 0.002$ (rejection). The Monte Carlo block estimates empirical power at $n = 20$ for detecting t_5 -distributed data — about 21 % — illustrating the test's modest power at small samples for mild departures.

Interpretation

A reporting sentence: “Shapiro-Wilk did not reject the Normality of the regression residuals ($W = 0.98$, $p = 0.52$), supporting the assumption underlying parametric inference. The complementary Q-Q plot showed close adherence to the diagonal, with no evidence of systematic deviation in the tails or at the centre. Parametric t -tests and ANOVA were therefore applied to these data without transformation. For comparison, residuals from a related model with skewed errors were rejected by the same test ($W = 0.83$, $p < 0.001$) and motivated a log transformation.” Always pair test with a Q-Q plot.

Practical Tips

- At very small samples ($n < 15$), Shapiro-Wilk has low power against most realistic departures and cannot reliably confirm Normality; absence of rejection at small samples is uninformative and should not be treated as evidence of Normality.
- At very large samples ($n > 1000$), the test rejects at trivial departures from Normality that are inconsequential for the validity of parametric inference; use Q-Q plots and effect-size summaries (skewness, kurtosis with their CIs) rather than the p -value alone in large samples.
- Graphical diagnosis via a Q-Q plot complements the numerical test and is often more informative; always plot the Q-Q before deciding whether departures are practically important. The combination of formal test and visual inspection is the modern recommendation.
- Normality assumptions in regression and ANOVA apply to the residuals, not to the raw outcome variable; testing Normality of the raw outcome is a recurring methodological error and provides no information about whether parametric inference is valid.
- When Normality fails substantively, consider transformations (log, Box-Cox, Yeo-Johnson), non-parametric alternatives (Wilcoxon, Mann-Whitney, Kruskal-Wallis), or robust methods (trimmed-mean inference, robust regression); the choice depends on the substantive question and the nature of the departure.
- For very heavy-tailed data, robust inference is often preferable to transformation; transformations can shift the substantive interpretation away from the original scale.

R Packages Used

Base R `shapiro.test()` for the canonical Shapiro-Wilk implementation up to $n = 5000$; `nortest::shapiro.test()` for an alternative interface; `nortest` for related Normality tests including Anderson-Darling, Cramér-von Mises, and Lilliefors; `moments::skewness()` and `moments::kurtosis()` for descriptive moment-based summaries; `car::qqPlot()` for enhanced Q-Q plot diagnostics complementing the formal test.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.

- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests